

File permissions in Linux

Project description

In this project, I reviewed and corrected file and directory permissions for a research team using Linux commands. I inspected existing permissions with `ls -la`, interpreted the 10-character permissions string, and applied the principle of least privilege by modifying permissions with `chmod`. I also corrected permissions on a hidden file and secured a directory so only the appropriate user could access it.

Check file and directory details

I used the following command to view all files, including hidden files, along with their permissions, owners, groups, and timestamps:

```
ls -la
```

This allowed me to identify which files had overly permissive access. Below is a summary of the initial permissions:

- **project_k.txt** — user: rw, group: rw, other: rw
- **project_m.txt** — user: rw, group: r, other: none
- **project_r.txt** — user: rw, group: rw, other: r
- **project_t.txt** — user: rw, group: rw, other: r
- **.project_x.txt** (hidden) — user: rw, group: w, other: none
- **drafts/** directory — user: rwx, group: x, other: none

From this list, several files violated the principle of least privilege because the group and/or other users had unnecessary read or write access.

Describe the permissions string

Linux permissions use a 10-character string such as:

```
-rw-rw-r--
```

The characters represent:

File type (- for file, d for directory)

2-4. **User permissions** (read, write, execute)

5-7. **Group permissions**

8-10. Other permissions

Example interpretation:

`-rw-rwr-`

`-` → regular file

`rw-` → user can read and write

`rw-` → group can read and write

`r-` → others can only read

Understanding this format allowed me to identify which files needed permission changes.

Change file permissions

I applied the principle of least privilege by removing unnecessary access from several files.

project_k.txt

Originally: `rw-rw-rw-`

Issue: The group and other users had write access.

Fix: `chmod go-rw project_k.txt`

project_r.txt

Originally: `rw-rw-rw-`

Issue: group had write access; others could read.

Fix: `chmod g-w,o-r project_r.txt`

project_t.txt

Originally: `rw-rw-r-`

Issue: same as `project_r.txt`.

Fix: `chmod g-w,o-r project_t.txt`

After these changes, only the file owner retained full access.

Change file permissions on a hidden file

The hidden file `.project_x.txt` had the following permissions:

- user: rw
- group: w
- other: none

Group users had write access but no read access, which is both unusual and insecure.

I corrected this by removing group write access: `chmod g-w .project_x.txt`

This ensures only the owner can modify the hidden file.

Change directory permissions

The `drafts/` directory originally allowed:

- user: rwx
- group: x
- other: none

To ensure only the intended user (researcher2) could access and modify the directory, I restricted group and other access: `chmod go-rwx drafts`

This prevents unauthorized users from entering or modifying the directory.

Summary

In this project, I inspected file and directory permissions using `ls -la`, interpreted permission strings, and applied the principle of least privilege to secure sensitive research files. I used `chmod` to remove unnecessary read and write access from multiple files, corrected permissions on a hidden file, and secured a directory so only the appropriate user could access it. These changes help protect confidential data and demonstrate effective Linux permission management.